

IBM Skills Build for University Engagements

AICTE-2026

Problem Statement No. 9 – AI Agent for Smart Farming Advice

The Challenge

Small-scale farmers often struggle to access timely, reliable, and localized agricultural information. AgroGenie AI addresses this challenge through an AI-powered Smart Farming Advisor built using IBM Cloud technologies and Retrieval-Augmented Generation (RAG). The system retrieves trusted information from weather services, agricultural knowledge bases, market price sources, and government resources to provide accurate and context-aware recommendations.

Farmers can interact with the assistant using **text, voice, or crop images** in their preferred language. They can ask questions such as: Which crop is suitable for this season? What is today's mandi price for tomatoes? Why are my crop leaves turning yellow? Which government schemes am I eligible for? AgroGenie AI combines specialized AI agents for crop recommendation, weather intelligence, disease advisory, market price analysis, government schemes, and agricultural knowledge retrieval. It also uses Farmer Memory and Explainability to provide personalized and transparent recommendations.

By integrating IBM watsonx.ai, llama models, IBM Cloud Object Storage, IBM Speech Services, MongoDB Atlas, and IBM Cloud Code Engine, the solution enables data-driven farming decisions, reduces agricultural risk, improves productivity, and helps bridge the knowledge gap for farmers.

Mandatory Technologies Used IBM Cloud IBM watsonx.ai llama Large Language Model Retrieval-Augmented Generation (RAG) IBM Cloud Object Storage IBM Speech to Text & Text to Speech